

Docker Lab

Cezary

What user are you logged in as by default?

As root.

If you start and then exit an interactive container, and then use the `docker run -it ubuntu:xenial /bin/bash` command again, is it the same container? How can you tell?

Not the same container, because different IDs:

```
root@d1d94f68a95a
```

```
root@1e8a8a1956df
```

(or run `docker ps -a` to check which independent containers got started).

Run the image you just built. Since we specified the default CMD, you can just do `docker run -it mypython:latest`. What do you observe?

It directly boots into the python command line rather than the system terminal (because of the CMD command in the Dockerfile).

Write and build a Dockerfile that installs the packages `fortune` and `fortunes-min` and runs the `fortune` executable (located in `/usr/games/fortune` after you install it). Note that you won't need to use the `-it` flags when you run the container as `fortune` doesn't need STDIN. Submit your Dockerfile with this lab. Hint: if you're having trouble writing your Dockerfile, try booting an interactive container and installing both packages. Translate what you did into a Dockerfile. How can you translate what you did interactively to a Dockerfile?

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Paste the output of your `docker images` command after questions 1 and 2.

```
czabo@cpc-2 dockerlab % docker run -it mypython:latest
```

```
Python 3.6.9 (default, Mar 10 2023, 16:46:00)
```

```
[GCC 8.4.0] on linux
```

```
Type "help", "copyright", "credits" or "license" for more information.
```

```
czabo@cpc-2 dockerlab % docker run fortune
```

```
Fine day to work off excess energy. Steal something heavy.
```

```
czabo@cpc-2 dockerlab % docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
fedora:latest	43b29f65a41e	282MB	69.2MB	U
fortune:latest	91b5b00c4c0e	149MB	47.7MB	U

hello-world:latest	5e2309035332	22.6kB	10.3kB	U
mypythn:latest	0ca840e7c9c6	188MB	56.9MB	U
ubuntu:latest	da6fc2be5478	180MB	44.5MB	U
ubuntu:xenial	1f1a2d56de1d	174MB	41.2MB	U

With httpd running in a detached container, run `/bin/bash` on the same container and paste the output of `ps aux`. Observe that there's very few processes running as compared to running `ps aux` on your VM. Why is this the case?

root@0f594c2cc5b7:/usr/local/apache2# ps aux

USER	PID	%CPU	%MEM	VSZ	RSS	TTY	STAT	START	TIME	COMMAND
root	1	0.0	0.0	8672	4688	?	Ss	16:33	0:00	httpd -DFOREGROUND
www-data	8	0.0	0.0	2001548	4872	?	SI	16:33	0:00	httpd -DFOREGROUND
www-data	9	0.0	0.0	2001492	4652	?	SI	16:33	0:00	httpd -DFOREGROUND
www-data	13	0.0	0.0	1935956	4660	?	SI	16:33	0:00	httpd -DFOREGROUND
www-data	92	0.0	0.0	2001484	4560	?	SI	16:35	0:00	httpd -DFOREGROUND
root	120	0.0	0.0	4176	3472	pts/0	Ss	16:40	0:00	/bin/bash
root	249	0.0	0.0	6016	3272	pts/0	R+	16:41	0:00	ps aux

I had to install "ps" first. This isn't a full operating system, but rather an isolated process on the host (we can only see the processes in the container).

Paste your Dockerfile for the Node.js web application

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Paste your docker-compose.yml file

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